CSA 0378-DATA STRUCTURES

14. LINEAR AND BINARY SEARCH

main.c

```c
#include <stdio.h>

int main()
{
    int a[100], n, key, i, found = 0;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Enter element to search: ");
    scanf("%d", &key);

    for(i = 0; i < n; i++)
    {
        if(a[i] == key)
        {
            found = 1;
            printf("Element found at position %d\n", i + 1);
            break;
        }
    }

    if(found == 0)
        printf("Element not found\n");

    return 0;
}
```

Output

```
Enter number of elements: 4
Enter elements:
10
20
30
40
Enter element to search: 20
Element found at position 2

=== Code Execution Successful ===
```

main.c

```c
#include <stdio.h>

int main()
{
    int a[100], n, key;
    int low, high, mid;
    int found = 0, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter sorted elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Enter element to search: ");
    scanf("%d", &key);

    low = 0;
    high = n - 1;

    while(low <= high)
    {
        mid = (low + high) / 2;

        if(a[mid] == key)
        {
            found = 1;
            printf("Element found at position %d\n", mid + 1);
            break;
        }
        else if(key < a[mid])
        {
            high = mid - 1;
        }
        else
```

Output

```
Enter number of elements: 4
Enter sorted elements:
10
20
30
40
Enter element to search: 20
Element found at position 2

=== Code Execution Successful ===
```

main.c

```c

    printf("Enter sorted elements:\n");
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Enter element to search: ");
    scanf("%d", &key);

    low = 0;
    high = n - 1;

    while(low <= high)
    {
        mid = (low + high) / 2;

        if(a[mid] == key)
        {
            found = 1;
            printf("Element found at position %d\n", mid + 1);
            break;
        }
        else if(key < a[mid])
        {
            high = mid - 1;
        }
        else
        {
            low = mid + 1;
        }
    }

    if(found == 0)
        printf("Element not found\n");

    return 0;
}
```

Output

```
Enter number of elements: 4
Enter sorted elements:
10
20
30
40
Enter element to search: 20
Element found at position 2

=== Code Execution Successful ===
```